

The empirical recurrence for OEIS A302462, recovered from its tail

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Abstract

OEIS A302462 records “Empirical recurrence of order 71” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 72$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 73$. Its last coefficient is $c_{71} = -336 \neq 0$, so no further tail satisfies anything shorter; any order-71 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A302462 is “Number of $4 \times n$ 0..1 arrays with every element equal to 0, 2, 3 or 5 horizontally, diagonally or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

8, 3, 55, 10, 233, 123, 1518, 2945, ...

The entry states:

Empirical recurrence of order 71 (see link above)

As of the “Last modified” line on the live entry (Apr 08 2018 10:04:38, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 72$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 71: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 71.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 144$ exact terms from $n = 2$ on, it returns order 71 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{71} c_i a(n-i),$$

c_1	5	c_{19}	−141501465	c_{37}	6892127313	c_{55}	−169331995
c_2	26	c_{20}	15825513	c_{38}	−9016898462	c_{56}	−773498753
c_3	−145	c_{21}	342836772	c_{39}	−5331000557	c_{57}	300869652
c_4	−348	c_{22}	−65760915	c_{40}	9278323227	c_{58}	261238615
c_5	2140	c_{23}	−731900458	c_{41}	2673541347	c_{59}	−187267678
c_6	3089	c_{24}	199319682	c_{42}	−7753302507	c_{60}	−72432648
c_7	−21062	c_{25}	1393367151	c_{43}	293459113	c_{61}	67232757
c_8	−19866	c_{26}	−503481775	c_{44}	4618675212	c_{62}	16241114
c_9	153572	c_{27}	−2386603660	c_{45}	−2338674795	c_{63}	−15422383
c_{10}	95859	c_{28}	1103765575	c_{46}	−867624367	c_{64}	−2966862
c_{11}	−877273	c_{29}	3693757633	c_{47}	2741608474	c_{65}	2285463
c_{12}	−347373	c_{30}	−2128186445	c_{48}	−2184761408	c_{66}	416497
c_{13}	4063747	c_{31}	−5161636258	c_{49}	−2019592381	c_{67}	−213685
c_{14}	902824	c_{32}	3635047351	c_{50}	3528114358	c_{68}	−37422
c_{15}	−15627249	c_{33}	6473203318	c_{51}	1102126757	c_{69}	11950
c_{16}	−1344558	c_{34}	−5534435802	c_{52}	−3079021520	c_{70}	1528
c_{17}	50779077	c_{35}	−7192443552	c_{53}	−339409621	c_{71}	−336
c_{18}	−1137843	c_{36}	7518806179	c_{54}	1808408890		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 73$, and it is the recurrence OEIS A302462 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{71} - c_1 x^{71-1} - \dots - c_{71}$. Its constant term is $-c_{71} = 336$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the

minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 71 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 71, so $\deg q = 71$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 71, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 25 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 71, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -336 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-71 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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